



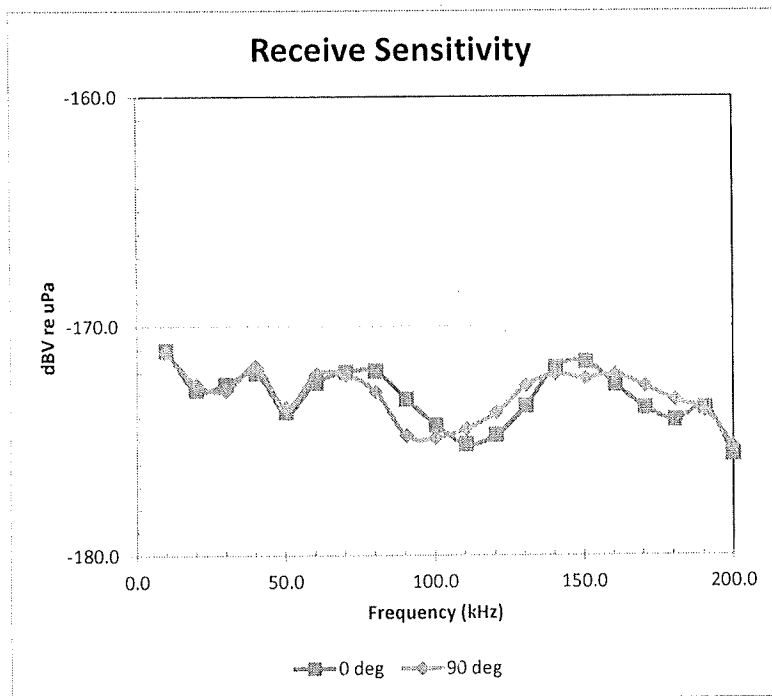
Certificate of Calibration

Ocean Sonics, Ltd.

Calibration Certificate Number: **C7708**

Test Result: 10 kHz to 100 kHz: -173.0 ± 1.9
10 kHz to 200 kHz: -173.3 ± 2.3

Model Number	SB35-ETH	Projector Manufacturer	Ocean Sonics
Serial Number	1249	Projector Model	TH2-SER-4F
Manufacture Date	11-Oct-2017	Projector Serial	2225
Measurement Date	13-12-2023	Measurement Distance	1 m
Certificate Date	13-12-2023	Output Level	130.1 dB re uPa @ 1 m
Sensitivity @ 26 Hz	-170.1 dB re V / uPa	Tone Burst	100.0 us / 300 ms
Case Type	Titanium	Reference Manufacturer	Ocean Sonics
Element Manufacturer	GeoSpectrum	Reference Model	RB9-ETH
Element Model	M24HF	Reference Serial	6465
Element Serial	I001092	Primary Calibration	2023-07-04
Preamp Model	04-300435-09	Preamp Manufacturer	Ocean Sonics
Calibrated By	D.Lee	Preamp Model	04-300451-01
Work Order Number	W45-135	Preamp Serial	Z88
Test Type	RX Sensitivity	Preamp Gain	36dB
Test Procedure	Complex RMS	ADC Manufacturer	Ocean Sonics
Test Location	Tank #3, 1 m	ADC Model Number	04-300427-01
Water Temperature	16 °C	ADC Serial Number	Z87



Frequency	Sensitivity [dBV re uPa]	
kHz	0 deg	90 deg
10.0	-171.0	-171.1
20.1	-172.8	-172.5
30.1	-172.5	-172.8
40.2	-172.0	-171.8
50.2	-173.7	-173.6
60.2	-172.5	-172.1
70.3	-172.0	-172.1
80.3	-172.0	-172.9
90.4	-173.2	-174.8
100.4	-174.3	-174.9
110.4	-175.2	-174.5
120.5	-174.8	-173.8
130.5	-173.5	-172.6
140.5	-171.8	-172.1
150.6	-171.5	-172.3
160.6	-172.6	-172.1
170.7	-173.6	-172.6
180.7	-174.1	-173.2
190.7	-173.6	-173.7
200.0	-175.6	-175.3